

Forced Magnetic Reconnection due to Time-Varying Resonant Magnetic Perturbations in Flowing Tokamak Plasmas

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Reply to Referees' Comments

The main criticisms of this paper is that it lacks focus on a clear physical problem, that it relies excessively on previously published research, and that it lack novelty. We have extensively rewritten the paper to address these concerns. Our reply to these criticisms are as follows:

1. We have now made it very clear that the physical problem that we are addressing is the response of a flowing tokamak plasma to a resonant magnetic perturbation that *varies in time*. We have changed the title of the paper to elucidate this point. We argue that, although the solution of the Taylor problem (i.e., the response of the plasma to a static RMP that is switched-on suddenly) may seem like an academic problem, once you understand the solution to this problem then you can immediately calculate the response of the plasma to a transient RMP of arbitrary time dependence. Of course, real tokamak plasmas are subject to many transient RMPs emanating from sawtooth crashes, ELMs, TAE modes, et cetera. Moreover, it is thought that some of these transient RMPs are responsible for triggering NTMs. However, no one has ever demonstrated this in detail. Our study is a first step in such a demonstration. Figs. 3 and 4 in the (new) paper give concrete examples showing how a knowledge of the solution of the Taylor problem allows us to determine the response of the plasma to fairly realistic transient perturbations.
2. Our Laplace transform analysis repurposes much of the Fourier transform analysis of Refs. 22 and 23 (in the new paper). Although we have attempted to cut down repetition of material from these papers (e.g., by starting from the linearized form of the three-field model), some repetition is necessary because we need to add new elements to the model (for example, the new layer parameter F_s), and we also have to recalculate all of the layer widths in terms of the Laplace transform variable (because that is how we know which response regime we are in). In order to be less repetitive, we have only described Reconnection Scenario 1 in any detail, assuming that the reader can glean the salient features of Scenarios 2–6 from Figs. 7 and 8 in the (new) paper. We have also now only performed a detailed comparison between our numerical solution and our analytic solutions for Reconnection Scenario 1, given that the other comparisons were fairly similar to this one.
3. With regards to the novelty of our results, the analysis of App. B has been generalized (from previously published material) to convert a Fourier transform problem into a Laplace transform problem. The analysis of App. C has been generalized to explain how to numerically calculate the new layer function, $F_s(\hat{g})$, in addition to the old one, $\hat{\Delta}_s(\hat{g})$. The analysis of App. D has been generalized to obtain analytic expressions for $F_s(\hat{g})$ in the various response regimes, as well as the layer widths as functions of $\hat{g}$. The closed-form expressions for the Taylor functions in the various different response regimes are all new, and obviously quite useful (especially as they are sufficiently simple that they can be calculated by means of a python script). The trajectories that the system takes through the various response regimes as time progresses, which are shown in Figs. 7 and 8 have also never been described before.

Our detailed comments to the referees' comments follow:

- Referee 1**
1. We have recalculated Figs. 1-4 in the new paper using $D = 2$, which is a more realistic value. The results are very similar to what we got with $D = 1.2$.
 2. Figs. 7 and 8 (in the new paper) are calculated on the basis of the layer widths as function of time, $\hat{\delta}(\hat{t})$, expressions that appear in Sect. IV.B. As is now described in the paper, the resonant layer width starts off with a comparatively large value than diminishes as time progresses. In Reconnection Scenario 1, the system is in the inertial regime as long as this regime has the

largest layer width, the system transitions to the viscous-inertial regime when the inertial layer width falls below the viscous-inertial width, and the system transitions to the viscous-resistive regime when the viscous-inertial layer width falls below the viscous-resistive layer width.

3. We have added more explanation to App. D to explain how layer widths are deduced.
4. The names of the various regimes are largely historical in nature. However, we have added more explanation as to what they denote in App. D.
5. A constant- ψ layer is a double layer, with an inner and an outer component in p . A non-constant- ψ layer lacks the inner component of a constant- ψ layer. Practically speaking, constant- ψ layers are characterized by $F_s = 1$, whereas non-constant- ψ layers are characterized by $|F_s| \ll 1$. Comments to this effect have been added to the paper.
6. There is no non-constant- ψ layer that corresponds to $D^2 p^2 \ll 1$ and $R \gg P p^2$ because this would lead to $k = -2$, and, hence, $\nu = \infty$, which is an impossible solution.
7. Fig. 6 in the paper is different to Fig. 2 of Ref. 22 because we are now insisting on equal layer thicknesses at regime boundaries, which was not necessarily the case on Ref. 22. Hence, the former figure is a correction to the latter. A comment has been added to this effect.

Referee 2 1. The seven response regimes in our paper are analogous to the seven response regimes of Ref. 22, but they are not exactly the same. We have had to generalize the analysis of Ref. 22 to express all quantities in terms of the Laplace transform variable, and also to calculate the new layer function, $F_s(\hat{g})$.

2. The expressions for the Taylor functions given in the seven response regimes are obtained after the performance of highly non-trivial integrations. These expressions are valuable because they are expressed in closed form and are sufficiently simple that they can be evaluated by a python script. In contrast, performing the contour integral numerically requires a sophisticated and complicated C++ program. Our expressions are all significantly modified by diamagnetic flows, which were completely ignored in all previous treatments of the Taylor problem, and are obviously important in real tokamaks.
3. The purpose of having the analytic method of calculating the Taylor function is that the numerical method is probably too slow to be used to analyze a lot of data.
4. The key issue addressed by the paper is how to determine the response of a realistic tokamak plasma to a time-varying RMP.
5. As has already been mentioned, analysis of the various analytic response regimes is not equivalent to that in Ref. 22 because we have been forced to add new elements: e.g., the calculation of $F_s(\hat{g})$.
6. The conclusions of the results is that the analytic approach gives rise to broadly similar results as the numerical approach, which is good, because it is a lot easier to deploy the analytic method to solving the problem.
7. The novelty of the results is that we have described a practical method for determining the response of a realistic tokamak plasma to a time-varying RMP of arbitrary form. The practical uses of this method are fairly obvious.

Referee 3 1. We have made a greater effort to highlight the novel features of the analysis.

2. We have added some comments in Sect. II.B that explain the main distinctions between the Laplace transform analysis performed in this paper and the Fourier transform analysis performed in previous papers.
3. We have added more language explaining the physical significance of the various reconnection regimes. Figs. 2–4 also further illustrate the physics of driven reconnection. Basically, if you just suddenly switch-on an RMP then it grows, starts to oscillate, and asymptotes to a steady-state. The various regimes mainly differ from one another because they are associated with different oscillation frequencies, and times to attain steady-state. However, if the RMP is transient then the time dependence is more interesting.

4. We have added an appendix that lists the major variables in the analysis and explains their significance.
5. We have added discussion to the final section that explains how the analysis of the paper could be used, for instance, to investigate the triggering of NTMs by transient RMPs in tokamaks.